Shrinking the Cross-Section: A Replication and a Machine-Learning Extension of Kozak, Nagel & Santosh (2020)

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Abstract

We replicate Kozak, Nagel & Santosh (2020, *Journal of Financial Economics*; hereafter KNS), "Shrinking the Cross-Section." KNS construct a robust stochastic discount factor (SDF) by imposing a prior that shrinks the contribution of low-variance principal components (PCs) of the candidate anomaly factors more aggressively, and argue that an L^2 (ridge) shrinkage of the PCs of the entire anomaly universe substantially outperforms sparse models that retain only a handful of factors. Our contributions are threefold. First, we restore the original Python port to a working state and reproduce the paper's four core results with a single command. Second, we carry out a machine-learning extension. Third, we add five further extensions and robustness analyses: the literal η=1 Pástor–Stambaugh benchmark, a generalized shrinkage curve that learns the shrinkage shape η, block-bootstrap confidence intervals, a look-ahead-free train-only PC rotation, and an FF25 preliminary analysis. The core findings reproduce cleanly: shrinkage is necessary, relative shrinkage is the key, and sparsity exists only in PC space. The machine-learning extension shows that endowing the shrinkage function with additional flexibility yields no statistically significant out-of-sample improvement, which confirms the paper's thesis that pricing information is diffuse and that linear relative shrinkage is already near-optimal. All results reproduce in roughly two minutes via `reproduce_all.py` under fixed random seeds. The replication code is publicly available at `https://github.com/Fomalhaut647/Replication-Shrinking-the-Cross-Section`.

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Contents

1 Introduction

The empirical asset-pricing literature has uncovered a large number of stock characteristics that predict the cross-section of expected returns. The dominant approach summarizes this evidence with a small number of characteristics-based factors (e.g. the Fama–French three- or five-factor models); that is, it seeks a characteristics-sparse SDF. KNS question this paradigm: when confronted with dozens or even thousands of candidate factors, a cross-sectional regression of average returns on factor covariances badly overfits the noise in the estimated means and performs poorly out of sample. Economically, the absence of near-arbitrage opportunities implies that any factor earning a substantial risk premium must be a major source of return covariance; pricing information should therefore be concentrated in the high-variance principal components. KNS accordingly impose a Bayesian prior that shrinks the SDF coefficients increasingly with the eigenvalue, and select the hyperparameters by maximizing the out-of-sample (OOS) cross-sectional R^2.

The paper advances three core claims, which form the backbone of our replication.

1. **Shrinkage is necessary, and relative shrinkage is the key.** The naive un-shrunk estimator collapses out of sample; equal (level) shrinkage in the spirit of Pástor–Stambaugh (η=1) performs far worse than relative shrinkage (η=2), which penalizes low-variance PCs more aggressively.

2. **There is almost no sparsity in characteristics space.** Forcing the SDF to retain only a few raw anomaly factors sharply degrades OOS performance, indicating little redundancy among the anomalies.

3. **Sparsity exists only in PC space.** A few high-variance PCs already approximate the SDF well, yet the pricing information is in fact spread over roughly twenty PCs rather than concentrated in the first few.

Our contribution. Building on lukaskoerber's Python port, we (i) fix porting bugs so that the pipeline runs under Python 3.14 and pandas 3; (ii) reproduce the four core results (Figures 1–4) within a unified pipeline; (iii) carry out a tiny-MLP nonlinear-shrinkage extension (Figure 5); and (iv) add five further extensions (Figures 6–8 together with a bootstrap analysis): the literal η=1 benchmark, the generalized shrinkage curve that learns η, block-bootstrap confidence intervals, a look-ahead-free rotation, and the FF25 preliminary analysis. The entire pipeline is CPU-only, minimal in its dependencies, and seed-fixed. All code is available at `https://github.com/Fomalhaut647/Replication-Shrinking-the-Cross-Section`.

2 Data

All inputs are portfolio-level returns or freely available Ken French data; we use no paid sources (no WRDS, CRSP, Compustat, or IBES).

- **50 anomaly long–short portfolios** (main dataset, `managed_portfolios_anom_d_50.csv`): daily, 1963-07 to 2017-12, with $T \approx 11{,}000$. Each column is the return of a zero-investment long–short portfolio sorted on one anomaly characteristic, augmented by the market excess return `rme`. This dataset underlies the paper's Section 4.2, Figures 3 and 4, and Table 1.

- **FF25** (25 size/BM portfolios) and the Fama–French factors: obtained from the Ken French data library and used in the paper's Section 4.1 as a low-dimensional, known-answer preliminary analysis (see extension R1, Section 6.5).

Following the original repository, the preprocessing first regresses out the market β of each column (de-marketing) and then rescales each series to the market volatility (de-volatilization). The factors F are thus market-orthogonalized long–short portfolios, and the SDF explains only the cross-sectional differences that remain beyond the market.

3 Methodology

3.1 The SDF and the shrinkage prior

We seek an SDF in the linear span of the factor returns,

$$M_t = 1 - b'\big(F_t - \mathbb{E}[F_t]\big), \qquad \mathbb{E}[M_t F_t] = 0 \;\Rightarrow\; b = \Sigma^{-1}\mathbb{E}[F_t], \tag{1}$$

where $\Sigma = \mathrm{Cov}(F_t)$. The naive sample estimator $\hat{b} = \bar{\Sigma}^{-1}\bar{\mu}$ badly overfits the noise in the estimated means in high dimensions. Rotating into principal-component coordinates, $\Sigma = QDQ'$ with $D = \mathrm{diag}(d_1, \ldots, d_N)$ and eigenvalues in decreasing order, and writing $P_t = Q'F_t$, KNS impose the prior family

$$\mu \sim \mathcal{N}\Big(0,\ \tfrac{\kappa^2}{\tau}\Sigma^{\eta}\Big), \qquad \tau = \mathrm{tr}(\Sigma), \tag{2}$$

in which κ controls the prior scale (the expected maximum squared Sharpe ratio) and η controls its shape. In PC space the prior variance of a PC's Sharpe ratio is proportional to $d_j^{\eta-1}$: η=0 makes the Sharpe ratios of low-variance PCs explode (near-arbitrage, implausible); η=1 gives all PCs the same expected Sharpe ratio (Pástor–Stambaugh); and η=2 makes high-variance PCs carry larger Sharpe ratios, which is economically plausible and keeps the implied portfolio weights bounded. KNS set η=2. A self-contained derivation of these statements is given in Appendix B.

3.2 The shrinkage estimator and relative shrinkage

With Σ treated as known and η=2, the posterior mean takes the ridge form

$$\hat{b} = (\Sigma + \gamma I)^{-1}\bar{\mu}, \qquad \gamma = \frac{\tau}{\kappa^2 T}, \tag{3}$$

whose PC-space expression reveals the essence of relative shrinkage:

$$\hat{b}_{P,j} = \underbrace{\frac{d_j}{d_j + \gamma}}_{\text{shrinkage weight } w_j} \cdot \underbrace{\frac{\bar{\mu}_{P,j}}{d_j}}_{\text{OLS coefficient}}. \tag{4}$$

PCs with a small eigenvalue d_j are shrunk harder (smaller w_j). The penalty parameters κ and γ are interchangeable through $\mathbb{E}[\mu'\Sigma^{-1}\mu]^{1/2} = \kappa$, which lends the penalty strength the economic reading of a root expected maximum squared Sharpe ratio.

3.3 The dual penalty and the OOS criterion

To permit sparsity, an L^1 penalty is added, yielding an elastic-net estimator:

$$\hat{b} = \arg\min_b\ (\bar{\mu} - \Sigma b)'\Sigma^{-1}(\bar{\mu} - \Sigma b) + \gamma_2\, b'b + \gamma_1 \|b\|_1. \tag{5}$$

Hyperparameters are chosen by K-fold contiguous-block cross-validation, and the OOS cross-sectional R^2 is defined as

$$R^2_{\text{oos}} = 1 - \frac{(\bar{\mu}_2 - \bar{\Sigma}_2\hat{b})'(\bar{\mu}_2 - \bar{\Sigma}_2\hat{b})}{\bar{\mu}_2'\bar{\mu}_2}, \tag{6}$$

where the subscript 2 denotes the held-out fold's sample moments. We implement this as a pure-function core, `scs_core.py`, and verify in `_selftest_core.py` that the elastic net reduces exactly to pure L^2 when γ_1=0 (error $\sim 10^{-15}$) and that the OOS curve reproduces the original repository's criterion.

3.4 The general-η unified shrinkage (used for R4 and EXT)

From the prior (2) we derive the PC-space posterior for arbitrary η, consistent with the KNS general prior and cross-checked independently (Appendix B):

$$\hat{b}_{P,j}(\eta,\gamma) = \frac{\bar{\mu}_{P,j}}{d_j + \gamma\, d_j^{2-\eta}} = w_j(\eta,\gamma)\cdot\frac{\bar{\mu}_{P,j}}{d_j}, \qquad w_j(\eta,\gamma) = \frac{d_j^{\eta-1}}{d_j^{\eta-1}+\gamma}. \tag{7}$$

At η=2 this recovers the ridge of (4); at η=1 the weight becomes the constant $1/(1+\gamma)$, an equal rescaling of the OLS coefficients (literal level shrinkage); and w_j increases monotonically in d_j. This unified form lets the literal P&S benchmark (R4), the generalized shrinkage curve (EXT), and the look-ahead removal (R2) share a single verified code path: at η=2 it reduces exactly to the original ridge (self-test error 8.4×10^{-15}).

4 Core replication results

All four core figures use the single criterion of eq. (6) under 3-fold contiguous-block CV. Table 1 summarizes the key numbers against the paper.

Table 1: Core replication numbers vs. the paper (50 anomalies, daily)

Quantity	This replication	Paper
OOS CV peak R^2 (Fig. 1)	0.252 @ κ^*=0.285	$\approx$ 0.30 @ $\kappa\approx$0.30
Best OOS at $\leq$ 10 nonzeros: PC vs char (Fig. 2)	0.233 vs 0.108	PC $\gg$ char
Order of PCs by $\lvert t\rvert$ (Fig. 3)	PC5, PC1, PC2, PC4, PC11...	PC5, PC1, PC2, PC4, PC11...
PC-sparse 4-factor OOS (Fig. 4)	0.209	$\approx$ 0.18
Char-sparse 4-factor OOS (Fig. 4)	0.020	low

4.1 Figure 1 — L2 model selection

In Figure 1 the horizontal axis is the prior strength κ (larger κ means a weaker penalty γ) and the vertical axis is the cross-sectional R^2. The in-sample curve (blue dashed) rises monotonically to 0.88: it fits the sample noise in the means and is spuriously high. The OOS curve (orange solid) is hump-shaped with an interior optimum, peaking at 0.252 at κ^*=0.285, the data-driven optimal degree of shrinkage. This is the central evidence that shrinkage is necessary. The gold dash-dotted line is the literal η=1 level shrinkage (extension R4, Section 6.2); its optimum reaches only 0.063, far below relative shrinkage.

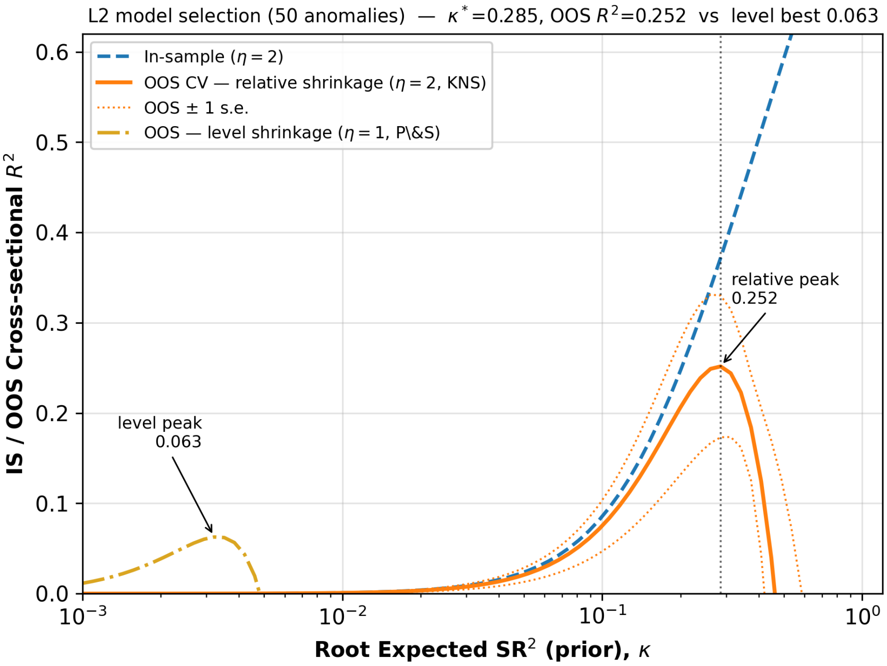

Figure 1: L2 model selection (paper Figure 4a). Relative shrinkage (η=2) peaks at OOS 0.252; literal level shrinkage (η=1, extension R4) tops out at only 0.063, and its optimal penalty is roughly two orders of magnitude stronger than that of relative shrinkage (the two humps sit on opposite sides of the axis), confirming that relative shrinkage is the key.

4.2 Figure 2 — L1–L2 dual-penalty contours

Figure 2 is a two-panel heatmap: the horizontal axis is κ (L^2 strength), the vertical axis is the number of nonzero coefficients (L^1 sparsity, on a log scale), and color encodes the OOS R^2. For the raw 50 anomalies (panel a) the high-scoring region sits at a high nonzero count and can hardly be made sparse; for their principal components (panel b) it extends down to very few nonzeros. At ≤ 10 nonzeros the best OOS in PC space is 0.233, against only 0.108 in characteristics space. This is the direct evidence that sparsity exists only in PC space.

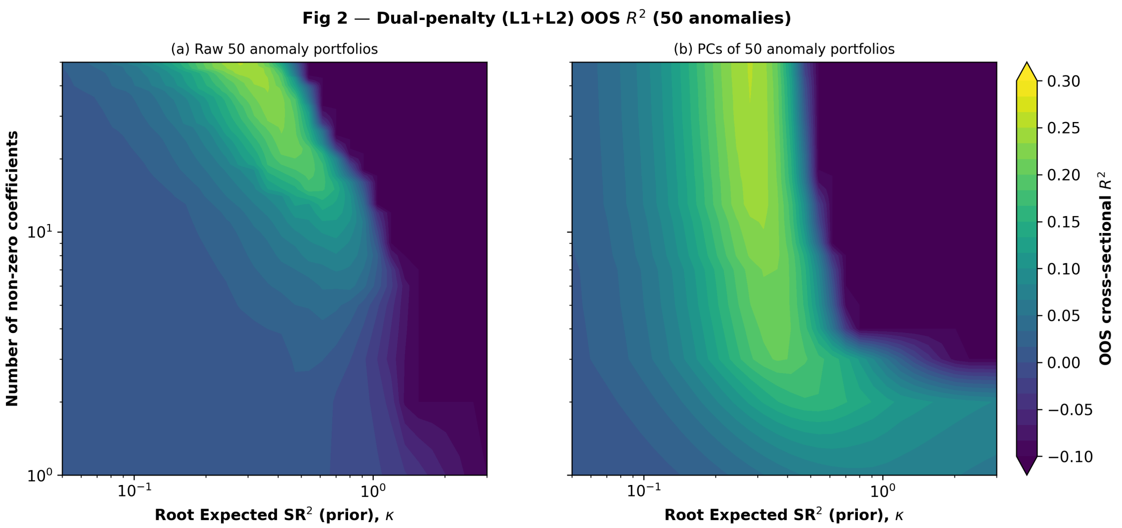

Figure 2: L1–L2 dual-penalty OOS R^2 contours (paper Figure 3a/3b). Left: raw characteristics; right: principal components. The high-scoring region extends to very few nonzeros in PC space but not in characteristics space.

4.3 Figure 3 — distribution of SDF coefficients in PC space

At the optimal κ^* we rotate the SDF coefficients into PC space (eq. 4); Figure 3 reports the $|t|$ statistic of each PC. The largest, in order, are PC5, PC1, PC2, PC4, PC11, PC15, PC10..., matching the paper's Table 1b almost item by item. The large loadings tilt toward high-variance PCs, but mid-variance PCs such as PC10, PC11, and PC15 also carry sizable loadings: the information is spread over roughly twenty PCs rather than concentrated in the first few, which is precisely the basis on which the paper rejects characteristics-sparsity in favor of PC shrinkage.

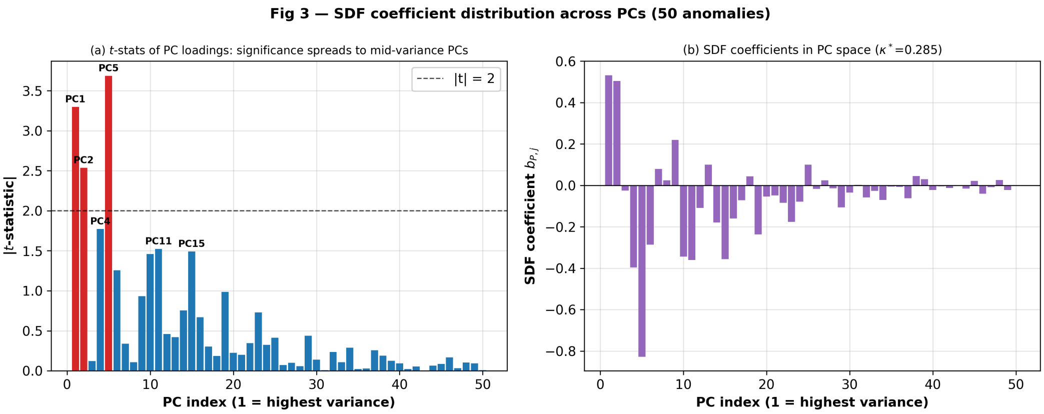

Figure 3: $|t|$ statistics and coefficients of the SDF in PC space at the optimal shrinkage (paper Table 1b).

4.4 Figure 4 — sparse vs. dense

In Figure 4 the horizontal axis is the number of factors entering the SDF and the vertical axis is the maximum OOS R^2 attainable at that sparsity level when sweeping all L^2 strengths (the ridge of the Figure 2 contours). The PC line rises quickly (2 PCs $\to$ 0.17, 4 $\to$ 0.21, 10 $\to$ 0.23, close to the maximum), whereas the characteristics line rises slowly (4 $\to$ 0.02, and roughly 50 are needed to catch up). The OOS annualized Sharpe ratio of the dense L^2 model (50 factors) is markedly higher than that of the sparse L^1 model (about 2 factors). This is the most direct OOS evidence that characteristics-sparsity fails while PC-sparsity holds.

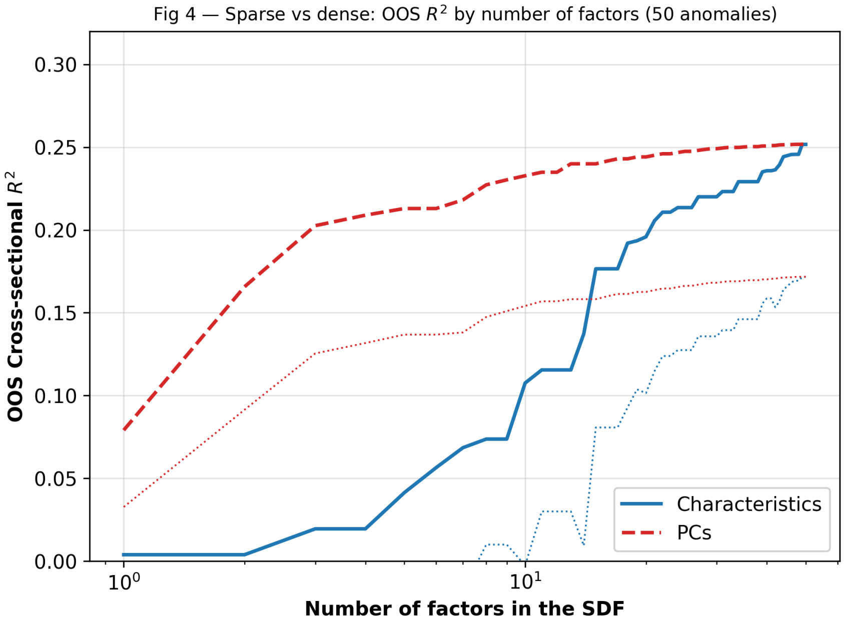

Figure 4: Sparse vs. dense OOS R^2 (paper Figure 4b). The PC line saturates quickly; the characteristics line rises slowly.

5 Machine-learning extension: a tiny MLP for nonlinear adaptive shrinkage

In PC space the paper's linear estimator is equivalent to a fixed shrinkage curve $w(d) = d/(d + \gamma)$. As a first machine-learning increment we replace it with an adaptive penalty γ_j learned by a tiny shallow MLP (one hidden layer, 16 neurons, with $\log d_j$ as the sole input, trained by SciPy L-BFGS with numerically verified analytic gradients), evaluated under exactly the same 3-fold OOS criterion as the four figures. The test block is fully held out, and the MLP is trained on an inner fit/validation split of the training block only, so there is no data leakage.

The MLP does not outperform the linear benchmark: the linear ridge attains OOS $R^2 \approx 0.22$ against the tiny MLP's ≈ 0.07 (Figure 5). This is not a tuning failure but a substantive finding. SDF coefficients are dominated by noise in the means and pricing information is diffuse, so what is required is strongly structured shrinkage that cannot be over-optimized in sample; granting the model additional nonlinear flexibility merely leads it to under-shrink and harms OOS performance. We did not tune the result in the MLP's favor. Section 6 reinforces this conclusion from the complementary, interpretable-knob angle. We note that the MLP implemented here is small enough that its expressiveness and denoising capacity are limited; we leave to future work the exploration of richer machine-learning architectures, which we did not pursue here owing to limited computational resources.

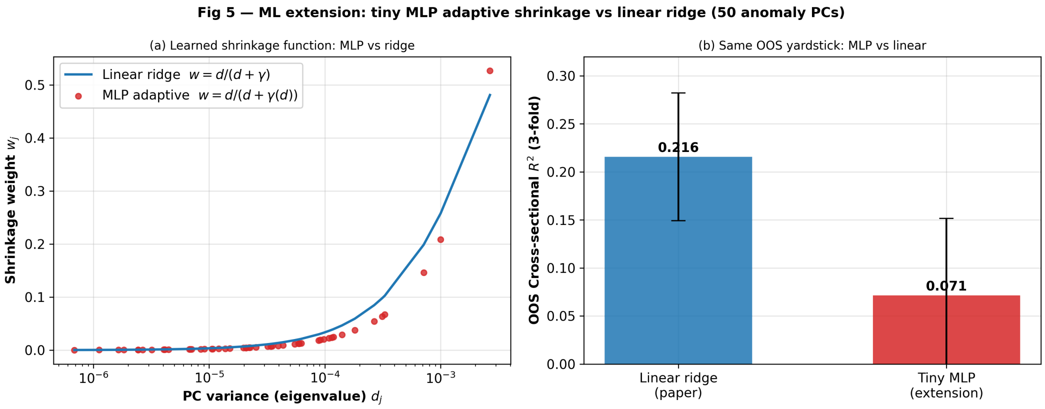

Figure 5: Tiny-MLP adaptive shrinkage vs. linear ridge. Left: the MLP's learned curve roughly tracks ridge; right: the OOS R^2 comparison, with the MLP clearly no better than the linear estimator.

6 Extensions and robustness analyses

This section presents our own work beyond the replication: five items, all CPU-only, seed-fixed, and folded into the one-click reproduction pipeline. Further implementation details are collected in Appendix C.

6.1 EXT — generalized shrinkage curve: learning the shrinkage shape η

The paper fixes η=2. We instead treat η as a learnable shrinkage shape and jointly search (η, γ) under exactly the same 3-fold OOS criterion as the four figures, asking whether additional flexibility in the shrinkage curve can beat η=2. The shrinkage weight is $w_j(\eta, \gamma)$ from eq. (7), itself a monotone shrinkage curve; each fold uses a train-only rotation, so there is no look-ahead.

Figure 6(a) plots the best OOS over γ against η: it rises steeply from η=0 (near-arbitrage, collapse), forms a broad plateau over $\eta \in [3, 4]$ (peak ≈ 0.29), and declines gently for η>4 as shrinkage becomes overly aggressive. The learned $\eta^* \approx 3.6$ gives OOS 0.292, a gain of +0.04 over the paper's η=2 value of 0.252. Figure 6(b) shows that the learned curve at η^* is steeper than at η=2: the data prefer stronger relative shrinkage of low-variance PCs.

This +0.04 gain, however, is not statistically significant (Section 6.3; the 95% CI of Δ contains zero). Its interpretation is twofold. First, learning the shrinkage shape yields no significant OOS improvement, so linear relative shrinkage is already near-optimal, consistent with the MLP result of Figure 5. Second, to the extent the data lean anywhere, they lean toward η>2—that is, toward stronger relative shrinkage—which aligns with and reinforces the paper's economic argument ($\eta \geq 2$, shrinking low-variance PCs harder) rather than contradicting it. The paper selects η=2 as the smallest value that keeps portfolio weights bounded; our result shows that η=2 sits on the left shoulder of a flat OOS plateau and is a robust, reasonable choice.

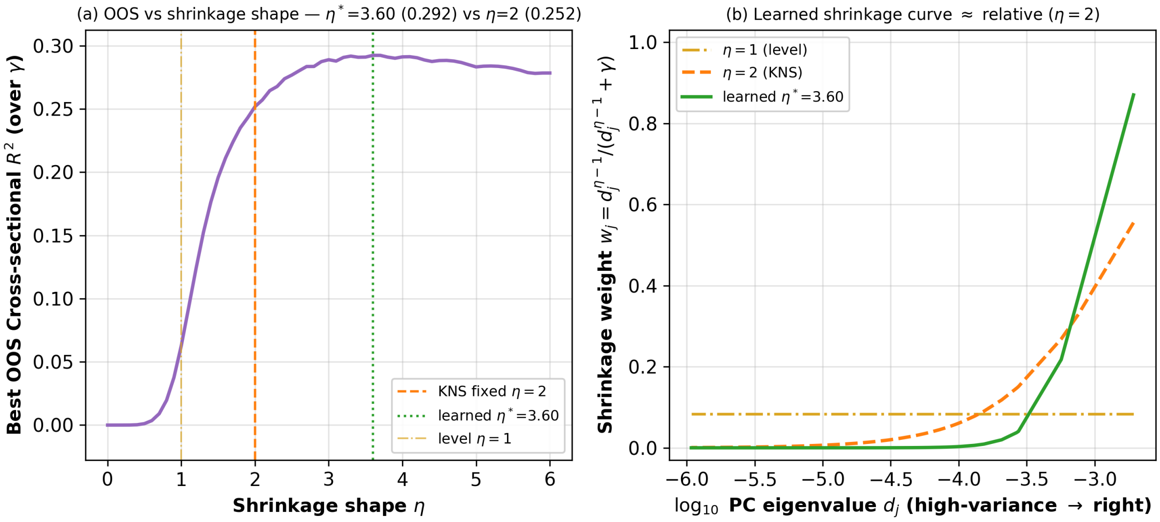

Figure 6: Generalized shrinkage curve (EXT). (a) Best OOS at each η: rise, broad plateau ($\eta^*{\approx}3.6$), gentle decline; (b) the learned curve $w(d)$ is steeper than at η=2, but the two are not significantly different out of sample (see R3).

6.2 R4 — the literal η=1 Pástor–Stambaugh benchmark

Pástor–Stambaugh (2000) corresponds to η=1 level (equal) shrinkage. Using the literal η=1 posterior $\hat{b}_{P,j} = \bar{\mu}_{P,j}/(d_j(1+\gamma))$ from eq. (7), we replace the original repository's level-shrinkage reparametrization and overlay the result on Figure 1. Because the natural penalty scales of level and relative shrinkage differ by about two orders of magnitude—their optimal humps sit on opposite sides of the κ axis, exactly as in the paper's remark that the x-axis no longer represents κ under η=1—we sweep level shrinkage over its own penalty range to its optimum. The best value is only $R^2_{\text{oos}} = 0.063$, about one quarter of relative shrinkage's 0.252. This treatment is more principled than the original repository's reparametrized version (0.038), and the qualitative conclusion is unchanged: equal shrinkage is far worse than relative shrinkage.

6.3 R3 — block-bootstrap confidence intervals

We attach 95% confidence intervals to the key OOS numbers via a leakage-free circular block bootstrap (block length 21 trading days, which preserves serial correlation). A methodological lesson is worth recording: naively resampling the whole series and then splitting into CV folds lets the same original observation land in both the train and the test fold, inducing leakage (which empirically inflates the OOS to ~ 0.8). The correct procedure is to resample independently within each fold's train and test windows, which are disjoint and hence leakage-free. The results are reported in Table 2:

- η=2 relative shrinkage gives OOS = 0.252, with a 95% CI of [0.014, 0.377], significantly above zero—shrinkage itself genuinely works.
- The EXT gain $\Delta = R^2(\eta^*{=}3.6) - R^2(\eta{=}2)$ has a 95% CI of $[-0.080, +0.109]$, which contains zero and is therefore not significant—confirming that learning the shrinkage shape yields no significant improvement and that η=2 is near-optimal.

The intervals are wide, because estimating cross-sectional means is precisely the most fragile step the paper emphasizes; they honestly reflect the intrinsic uncertainty of out-of-sample pricing.

Table 2: Block-bootstrap confidence intervals (50 anomalies, n_{boot}=300, block length 21)

Model	Point OOS R^2	95% CI
η=1 (level)	0.063	[−0.001, 0.219]
η=2 (relative, KNS)	0.252	[0.014, 0.377]
η^*=3.6 (learned)	0.292	[0.034, 0.411]
Gain $\Delta = R^2(\eta^*) - R^2(\eta{=}2)$	+0.040	[−0.080, +0.109] (contains 0)

6.4 R2 — removing look-ahead: a train-only PC rotation

The replication backbone rotates factors into PC space using the eigenvectors Q of the full-sample covariance, so the training block effectively sees a Q estimated partly from the test block—a mild look-ahead. We quantify it rigorously, and the conclusion splits into two cases (Figure 7):

- **Pure L^2 ridge** (the Figure 1 backbone and the L^2 envelope of Figure 4): under the isotropic penalty γI, ridge is rotation-equivariant, so the OOS R^2 is exactly independent of the choice of Q. The difference between full-sample Q and train-only Q is 3.3×10^{-15}, i.e. zero at machine precision.
- **PC-sparse** (L^1 selection of PCs, which is rotation-dependent): comparing the global maximum OOS of the two versions, train-only Q gives 0.269, no lower than full-sample Q's 0.252. The look-ahead does not inflate the backbone PC results; if anything they rise slightly once it is removed.

Moreover, the generalized shrinkage curve of Section 6.1 (and the η=1 curve of Figure 1) already use train-only Q by default, so they carry no such look-ahead. The paper's PC conclusions are therefore robust to removing it.

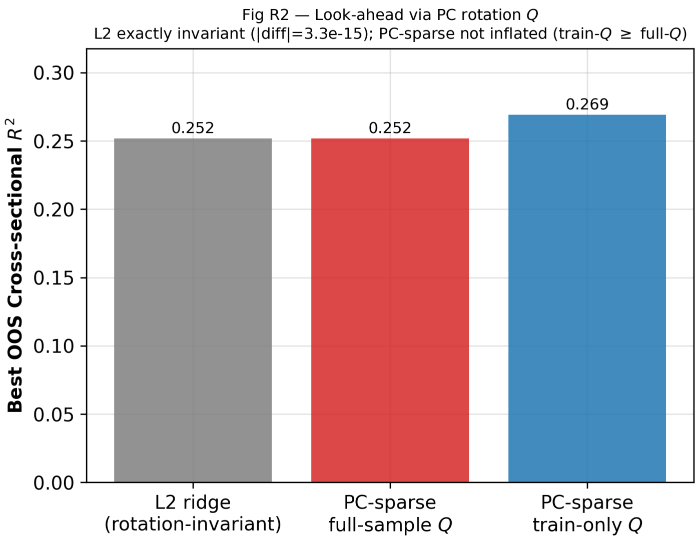

Figure 7: Look-ahead via the PC rotation Q (R2). Pure L^2 is rotation-invariant (difference $\sim 10^{-15}$); the PC-sparse global optimum under train-only Q is no lower than under full-sample Q, so the backbone results are not inflated.

6.5 R1 — FF25 preliminary analysis (a negative control)

The paper's Section 4.1 uses FF25—a low-dimensional, known-answer setting—to check that the method recovers an SMB/HML-like few-factor SDF. We run the exact same KNS pipeline as for the 50 anomalies on FF25 (over the same 1963–2017 window) and find (Figure 8):

- (a) FF25 is lower-dimensional: the top three PCs account for 65% of the variance, whereas the 50 anomalies are more spread out. This is consistent with the paper's use of FF25 as a low-dimensional warm-up.
- (b) After de-marketing, however, the residual cross-section has OOS $R^2 \approx 0$ (with large standard errors): the size and value residual premia are strongly time-varying and almost unpredictable out of sample under a 3-fold time split.

This is a valuable negative control: the same pipeline honestly returns about zero when there is no stable signal, which argues that the 0.25 obtained on the 50 anomalies is not a mechanical artifact. It also shows that the OOS value of the KNS method appears specifically in the high-dimensional anomaly cross-section, not in the classic low-dimensional FF25—in line with the paper's premise that the multidimensional challenge bites only in high dimensions.

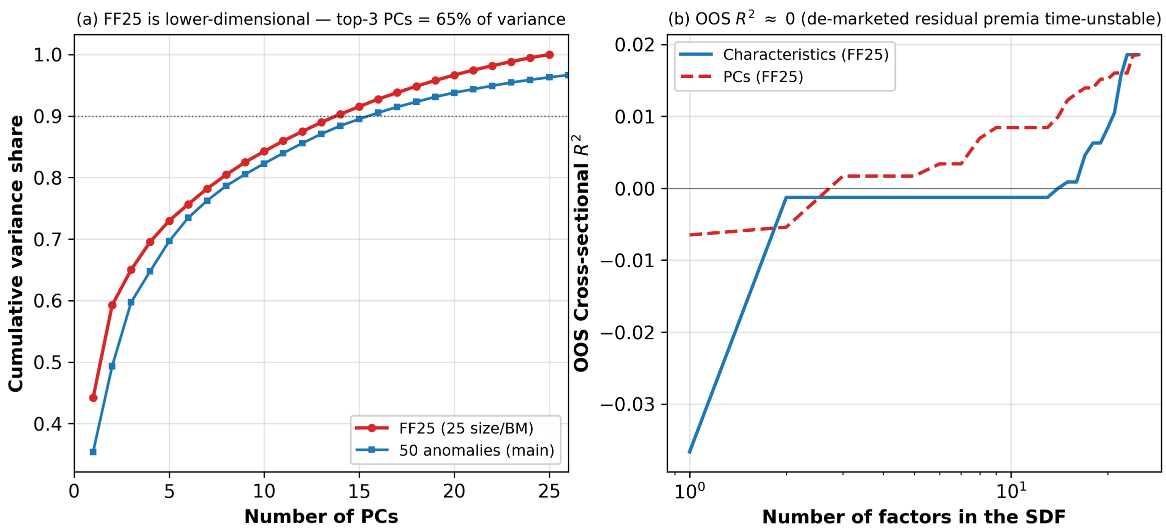

Figure 8: FF25 preliminary analysis (R1). (a) FF25 is lower-dimensional than the 50 anomalies (cumulative variance); (b) the de-marketed residual cross-section has OOS $R^2 \approx 0$ (time-varying premia), serving as a negative control.

7 Limitations

- **Full-sample β in de-marketing.** This follows the paper's own approach; it introduces a mild look-ahead, but it is very stable given the large daily sample.
- **The PC coefficients of Figure 3** are an in-sample descriptive quantity at the optimal κ^*, whose PC rotation uses the full-sample Q; this is descriptive and inherent to the exercise.
- **The bootstrap intervals are wide.** With only 3 folds and large noise in estimating cross-sectional means, the width is an honest reflection rather than a defect.
- $\eta^* \approx 3.6$ **lies near the upper end of the search.** We extended the grid to $\eta \in [0, 6]$ to confirm both the plateau and the subsequent decline, so η^* is not a grid-boundary artifact; but the plateau is flat, the precise value of η^* is not sharply identified, and its gain over η=2 is correspondingly insignificant.
- **No paid data.** We cannot reproduce the paper's ultra-high-dimensional dataset of thousands of characteristic interactions; the main 50-anomaly set nonetheless supports all qualitative conclusions.

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A The Stochastic Discount Factor: A Brief Review

This appendix provides a self-contained review of the asset-pricing framework underlying the paper, so that the derivations in Appendix B are readable without recourse to external references.

The fundamental pricing equation. The no-arbitrage foundation of modern asset pricing is the existence of a strictly positive random variable M_t, the stochastic discount factor (SDF, also called the pricing kernel), such that the price of any traded payoff equals the expectation of the discounted payoff. For excess returns R_t of zero-cost portfolios this specializes to

$$\mathbb{E}_{t-1}[M_t R_t] = 0, \tag{8}$$

since a zero-cost position has zero price. Equation (8) states that, after discounting by M_t, no traded excess return is expected to earn anything: all priced variation is absorbed into the kernel.

From the kernel to factor models. Following Hansen and Jagannathan (1991), one can always find an SDF inside the linear span of the returns under consideration. Writing the candidate factors (themselves returns on zero-investment characteristic-sorted portfolios) as the $N \times 1$ vector F_t, we posit

$$M_t = 1 - b'\big(F_t - \mathbb{E}[F_t]\big), \tag{9}$$

where b is the $N \times 1$ vector of SDF loadings. The constant is normalized so that $\mathbb{E}[M_t] = 1$. The SDF loadings b are the central object of estimation: a factor with a nonzero loading is a priced source of risk, i.e. it contributes to the variation of the kernel.

Economic meaning of b. Two interpretations of b recur throughout the paper. First, by the pricing equation (8)–(9) (derived in Appendix B), $b = \Sigma^{-1}\mathbb{E}[F_t]$: the loadings are the coefficients of a cross-sectional regression of expected returns on factor covariances. Second, the SDF loadings coincide, up to scale, with the weights of the mean-variance efficient (MVE) portfolio of the factors. The difficulty of estimating b when N is large is therefore the same difficulty, long known in the portfolio-choice literature, of estimating MVE weights when the number of assets is large: the sample mean vector $\bar{\mu}$ is estimated imprecisely, and inverting Σ amplifies that noise, especially along low-variance directions.

Maximum Sharpe ratio and near-arbitrage. For an SDF of the form (9), the squared maximum Sharpe ratio attainable from the factors equals $\mathbb{E}[F_t]'\Sigma^{-1}\mathbb{E}[F_t] = \mu'\Sigma^{-1}\mu$. A prior belief that this quantity cannot be implausibly large is precisely a belief that near-arbitrage opportunities do not exist. This is the economic content that the KNS prior formalizes, and it is what ties the first moments of returns (the means μ) to their second moments (the covariance Σ): high Sharpe ratios are admissible only along high-variance directions.

B Theoretical Derivations

This appendix derives the equations used in Sections 3.1–3.4. Throughout, $\Sigma = QDQ'$ is the eigendecomposition of the factor covariance, $D = \mathrm{diag}(d_1, \ldots, d_N)$ with $d_1 \geq \cdots \geq d_N > 0$, $P_t = Q'F_t$ are the principal-component (PC) portfolios, and a P subscript denotes the PC-space counterpart of a quantity (e.g. $\mu_P = Q'\mu$, $\bar{\mu}_P = Q'\bar{\mu}$). Functions of Σ are understood spectrally: $\Sigma^a = QD^aQ'$.

B.1 The pricing equation and the naive estimator

We first derive $b = \Sigma^{-1}\mathbb{E}[F_t]$ from (8)–(9). Substituting the linear SDF into the unconditional pricing equation $\mathbb{E}[M_t F_t] = 0$,

$$\mathbb{E}[M_t F_t] = \mathbb{E}[F_t] - \mathbb{E}\big[F_t\,(F_t - \mathbb{E}[F_t])'\big]\,b = 0.$$

The middle expectation is the covariance matrix, because

$$\mathbb{E}\big[F_t(F_t - \mathbb{E}[F_t])'\big] = \mathbb{E}\big[(F_t - \mathbb{E}[F_t])(F_t - \mathbb{E}[F_t])'\big] + \mathbb{E}[F_t]\,\mathbb{E}\big[(F_t - \mathbb{E}[F_t])'\big] = \Sigma + 0 = \Sigma.$$

Hence $\mathbb{E}[F_t] = \Sigma b$, i.e.

$$b = \Sigma^{-1}\mathbb{E}[F_t]. \tag{10}$$

Replacing population moments by sample moments $\bar{\mu} = \frac{1}{T}\sum_t F_t$ and $\bar{\Sigma}$ yields the naive estimator $\hat{b}_{\text{ols}} = \bar{\Sigma}^{-1}\bar{\mu}$. In PC space its components are $\hat{b}^{\text{ols}}_{P,j} = \bar{\mu}_{P,j}/d_j$; since $\text{var}(\hat{b}_{P,j}) \propto d_j^{-1}$, the estimator has explosive sampling uncertainty along the smallest-eigenvalue directions, which is the source of its OOS fragility.

B.2 The prior in PC space and the role of η

The prior (2), $\mu \sim \mathcal{N}\big(0, \frac{\kappa^2}{\tau}\Sigma^\eta\big)$ with $\tau = \text{tr}(\Sigma)$, rotates into PC space as

$$\mu_P \sim \mathcal{N}\Big(0,\ \tfrac{\kappa^2}{\tau}D^\eta\Big), \qquad \text{so} \qquad D^{-1/2}\mu_P \sim \mathcal{N}\Big(0,\ \tfrac{\kappa^2}{\tau}D^{\eta-1}\Big), \tag{11}$$

where $D^{-1/2}\mu_P$ is the vector of prior Sharpe ratios of the PCs. The prior variance of PC j's Sharpe ratio is therefore proportional to $d_j^{\eta-1}$, which delivers the economic reading of η:

- $\eta = 0$: the Sharpe-ratio variance is $\propto d_j^{-1}$ and explodes for small d_j; the prior admits near-arbitrage and is economically implausible.
- $\eta = 1$: the Sharpe-ratio variance is constant across PCs; all PCs are a priori equally rewarded (Pástor–Stambaugh). This is the limiting case of level shrinkage.
- $\eta = 2$: the Sharpe-ratio variance is $\propto d_j$, so high-variance PCs are expected to carry the larger Sharpe ratios—the economically motivated case, and the one that keeps portfolio weights bounded (see Appendix B.5).

B.3 The posterior mean for general η

Treat Σ (hence D and Q) as known, which is the maintained assumption of the paper, justified by the high precision of daily covariance estimates. With a Gaussian likelihood the sample mean satisfies $\bar{\mu} \mid \mu \sim \mathcal{N}(\mu, \Sigma/T)$. Combining this with the conjugate prior $\mu \sim \mathcal{N}(0, \Sigma_0)$, $\Sigma_0 = \frac{\kappa^2}{\tau}\Sigma^\eta$, the standard normal–normal update gives the posterior mean of μ,

$$\hat{\mu} = \big(\Sigma_0^{-1} + T\Sigma^{-1}\big)^{-1}\big(\Sigma_0^{-1}\cdot 0 + T\Sigma^{-1}\bar{\mu}\big) = \big(\Sigma_0^{-1} + T\Sigma^{-1}\big)^{-1}T\Sigma^{-1}\bar{\mu}. \tag{12}$$

Substituting $\Sigma_0^{-1} = \frac{\tau}{\kappa^2}\Sigma^{-\eta}$ and factoring Σ^{-1} on the left,

$$\Sigma_0^{-1} + T\Sigma^{-1} = \tfrac{\tau}{\kappa^2}\Sigma^{-\eta} + T\Sigma^{-1} = \Sigma^{-1}\Big(\tfrac{\tau}{\kappa^2}\Sigma^{1-\eta} + TI\Big),$$

so that, with $\gamma \equiv \tau/(\kappa^2 T)$,

$$\hat{\mu} = \Big(\tfrac{\tau}{\kappa^2}\Sigma^{1-\eta} + TI\Big)^{-1}\Sigma\cdot T\Sigma^{-1}\bar{\mu} = \Big(\tfrac{\tau}{\kappa^2 T}\Sigma^{1-\eta} + I\Big)^{-1}\bar{\mu} = \big(\gamma\,\Sigma^{1-\eta} + I\big)^{-1}\bar{\mu}.$$

Because $\hat{b} = \Sigma^{-1}\hat{\mu}$ by (10),

$$\hat{b} = \Sigma^{-1}\big(\gamma\,\Sigma^{1-\eta} + I\big)^{-1}\bar{\mu} = \big(\Sigma + \gamma\,\Sigma^{2-\eta}\big)^{-1}\bar{\mu}. \tag{13}$$

In PC space, eq. (13) is diagonal:

$$\hat{b}_{P,j}(\eta,\gamma) = \frac{\bar{\mu}_{P,j}}{d_j + \gamma\, d_j^{2-\eta}} = \underbrace{\frac{d_j^{\eta-1}}{d_j^{\eta-1} + \gamma}}_{w_j(\eta,\gamma)} \cdot \frac{\bar{\mu}_{P,j}}{d_j}, \tag{14}$$

where the second equality divides numerator and denominator by $d_j^{2-\eta}$. This is exactly eq. (7). The weight $w_j \in (0,1)$ is increasing in d_j for every $\gamma > 0$ whenever $\eta > 1$, so low-variance PCs are shrunk harder; at $\eta = 1$ the weight collapses to the constant $1/(1+\gamma)$, reproducing pure level shrinkage.

The KNS ridge as the special case $\eta = 2$. Setting $\eta = 2$ in (13) gives $\Sigma^{2-\eta} = \Sigma^0 = I$ and therefore the ridge form (3),

$$\hat{b} = (\Sigma + \gamma I)^{-1}\bar{\mu}, \qquad \hat{b}_{P,j} = \frac{d_j}{d_j + \gamma} \cdot \frac{\bar{\mu}_{P,j}}{d_j},$$

which is eq. (4). Comparison with the OLS coefficient $\bar{\mu}_{P,j}/d_j$ makes the shrinkage transparent: every PC coefficient is multiplied by $d_j/(d_j + \gamma) < 1$, and the attenuation is strongest exactly where the eigenvalue—and thus the contribution to SDF volatility—is smallest.

B.4 The economic scale of the penalty: κ

We verify the identity $\mathbb{E}[\mu'\Sigma^{-1}\mu]^{1/2} = \kappa$ used to give γ an economic reading. Under the $\eta = 2$ prior, working in PC space where $\mu'\Sigma^{-1}\mu = \mu_P' D^{-1}\mu_P$ and $\mu_P \sim \mathcal{N}(0, \frac{\kappa^2}{\tau}D^2)$,

$$\mathbb{E}[\mu'\Sigma^{-1}\mu] = \mathbb{E}[\mu_P' D^{-1}\mu_P] = \operatorname{tr}\Big(D^{-1}\cdot \tfrac{\kappa^2}{\tau}D^2\Big) = \tfrac{\kappa^2}{\tau}\operatorname{tr}(D) = \tfrac{\kappa^2}{\tau}\,\tau = \kappa^2.$$

Hence κ is the root expected maximum squared Sharpe ratio implied by the prior, and through $\gamma = \tau/(\kappa^2 T)$ a small κ (a belief that high Sharpe ratios are implausible) maps into a large γ (strong shrinkage).

B.5 Boundedness of portfolio weights and the choice $\eta \geq 2$

The requirement that the implied portfolio be economically sensible is that $\mathbb{E}[b'b]$ remain finite. With $b = \Sigma^{-1}\mu$ we have $b'b = \mu'\Sigma^{-2}\mu = \mu_P' D^{-2}\mu_P$, so under the prior (11),

$$\mathbb{E}[b'b] = \operatorname{tr}\Big(D^{-2}\cdot \tfrac{\kappa^2}{\tau}D^{\eta}\Big) = \frac{\kappa^2}{\tau}\sum_{i=1}^{N} d_i^{\eta-2}. \tag{15}$$

Because the smallest eigenvalues d_i are extremely close to zero, the term $d_i^{\eta-2}$ diverges whenever $\eta < 2$: the prior then places unboundedly large bets on the lowest-variance PCs. Setting $\eta \geq 2$ keeps (15) finite, and $\eta = 2$ is the smallest (hence least restrictive, "flattest") such value, which is why KNS adopt it.

B.6 Penalized representations and the dual penalty

The Bayesian estimator (3) also arises from two penalized objectives that connect it to machine learning. Maximizing the model cross-sectional fit subject to a penalty on the implied maximum squared Sharpe ratio $\gamma\, b'\Sigma b$,

$$\hat{b} = \arg\min_b\ (\bar{\mu} - \Sigma b)'(\bar{\mu} - \Sigma b) + \gamma\, b'\Sigma b, \tag{16}$$

has first-order condition $-2\Sigma(\bar{\mu} - \Sigma b) + 2\gamma\Sigma b = 0$, i.e. $\Sigma\bar{\mu} = \Sigma(\Sigma + \gamma I)b$, giving $\hat{b} = (\Sigma + \gamma I)^{-1}\bar{\mu}$, exactly (3). Equivalently, minimizing the Hansen–Jagannathan distance subject to an L^2 penalty $\gamma\, b'b$,

$$\hat{b} = \arg\min_b\ (\bar{\mu} - \Sigma b)'\Sigma^{-1}(\bar{\mu} - \Sigma b) + \gamma\, b'b, \tag{17}$$

yields the same solution. Adding an L^1 penalty to (17) to permit factor selection produces the elastic-net estimator of eq. (5),

$$\hat{b} = \arg\min_b\ (\bar{\mu} - \Sigma b)'\Sigma^{-1}(\bar{\mu} - \Sigma b) + \gamma_2\, b'b + \gamma_1\|b\|_1,$$

whose L^2 component carries the economic shrinkage and whose L^1 component sets some coefficients exactly to zero. Setting $\gamma_1 = 0$ recovers (17) exactly—the reduction verified numerically to $\sim 10^{-15}$ in `_selftest_core.py`.

The cross-validated OOS criterion. All hyperparameters are selected by maximizing the cross-sectional OOS R^2 of eq. (6). With the data partitioned into K contiguous blocks, for each candidate penalty the estimator $\hat{b}$ is fitted on $K-1$ blocks and scored on the withheld block via

$$R^2_{\text{oos}} = 1 - \frac{(\bar{\mu}_2 - \bar{\Sigma}_2\hat{b})'(\bar{\mu}_2 - \bar{\Sigma}_2\hat{b})}{\bar{\mu}_2'\bar{\mu}_2},$$

where subscript 2 denotes the withheld block's sample moments; the K fold scores are then averaged. We use $K = 3$, the paper's choice, which balances estimation uncertainty in $\hat{b}$ against the difficulty of estimating the held-out covariance $\bar{\Sigma}_2$ precisely. Contiguous blocks (rather than random folds) preserve the time-series structure and prevent leakage across train and test windows.

C Implementation and Technical Details

This appendix collects the implementation details of the replication and the extensions. The complete, runnable code is available at `https://github.com/Fomalhaut647/Replication-Shrinking-the-Cross-Section`.

Numerical core. The estimator is implemented as a set of pure functions in `scs_core.py` that map sample moments $(\bar{\mu}, \bar{\Sigma})$ and penalties to $\hat{b}$ and to the OOS R^2 of eq. (6). The general-η path of eq. (14) is the single code route shared by the ridge backbone, the η=1 benchmark (R4), the generalized shrinkage curve (EXT), and the look-ahead study (R2). Six numerical self-tests in `_selftest_core.py` guard the core, including: (i) the elastic net reduces to pure L^2 at γ_1=0 to within $\sim 10^{-15}$; (ii) the general-η path reduces to the original ridge at η=2 to within 8.4×10^{-15}; and (iii) the OOS-CV curve matches the original repository's criterion.

Data pipeline. `scs_data.py` loads the 50-anomaly daily file and the FF25 portfolios over the common 1963–2017 window, then de-markets each column by regressing out its full-sample market β and de-volatilizes it to the market volatility, exactly as in the original repository. The same routine prepares the FF25 inputs (`prepared_ff25`) so that R1 runs the identical pipeline on a different dataset.

Cross-validation. Penalties are selected by 3-fold contiguous-block CV. Each fold's training block supplies the moments and (for the PC-space analyses) the rotation Q; the held-out block is scored once via eq. (6). The dual-penalty sweep (`dual_penalty.py`) traces the (κ, nonzeros) grid that underlies the Figure 2 contours and the Figure 4 frontier, and exposes a train-only-Q option used by R2.

Tiny-MLP extension (Figure 5). The adaptive shrinkage of Section 5 replaces the fixed weight $w(d) = d/(d + \gamma)$ by a penalty γ_j emitted by a one-hidden-layer MLP (16 neurons, input $\log d_j$). The network is trained by SciPy L-BFGS with analytic gradients whose correctness is checked against finite differences. To preclude leakage, the test block is fully withheld; the MLP is fitted on an inner fit/validation split of the training block only, and is scored under the same 3-fold OOS criterion as the core figures.

Generalized shrinkage curve (Figure 7). `gen_shrinkage.py` jointly searches (η, γ) over $\eta \in [0, 6]$ using the weight $w_j(\eta, \gamma)$ of eq. (14), with a train-only rotation in every fold. For each η the best OOS over γ is recorded, producing the rise–plateau–decline profile of Figure 6(a).

Block bootstrap (R3). `bootstrap_ci.py` implements a circular block bootstrap with block length 21 trading days and $n_{\text{boot}} = 300$. The key design point is leakage avoidance: resampling is performed independently inside each fold's disjoint train and test windows, rather than resampling the full series before splitting (which would let one observation appear in both windows and spuriously inflate the OOS to ~ 0.8).

Look-ahead study (R2). `r2_lookahead.py` compares full-sample Q against train-only Q. For pure L^2 ridge the OOS is rotation-equivariant, so the two coincide to machine precision (3.3×10^{-15}); for PC-sparse selection the global-maximum OOS under train-only Q (0.269) is no lower than under full-sample Q (0.252).

FF25 negative control (Figure 6). `fig6_ff25_sparsity.py` runs the identical KNS pipeline on FF25, reporting the cumulative PC variance (top three PCs $\approx 65\%$) and the de-marketed residual OOS $R^2 \approx 0$, which serves as the negative control discussed in Section 6.5.

D Reproducibility and Environment

Hard constraints respected: CPU-only, with no GPU or CUDA; no paid data (only portfolio-level returns and freely available Ken French data); dependencies limited to `numpy`, `pandas`, `matplotlib`, and `scipy`; and fixed random seeds (`np.random.seed(0)` and `default_rng(0)`).

One-click reproduction (about 2 minutes, producing Figures 1–8 in `outputs/`):

```
uv sync                         # install dependencies
uv run python reproduce_all.py  # Fig. 1-8 in outputs/ (~2 min)
uv run python _selftest_core.py # 6 numerical self-tests
```

Environment: Python 3.14, numpy 2.4, pandas 3.0, matplotlib 3.10, scipy 1.17 (versions locked in `uv.lock` and `requirements.txt`). Code layout: `scs_core` (numerical core, including the unified general-η OOS-CV), `scs_data` (unified data pipeline, including `prepared_ff25`), `dual_penalty` (dual-penalty sweep, including the train-Q option), `fig1--4` (core figures), `mlp_sdf` (Figure 5), `gen_shrinkage` (Figure 7), `r2_lookahead` (Figure 8), `bootstrap_ci` (R3), and `fig6_ff25_sparsity` (Figure 6). The full repository, including this layout and the locked environment, is hosted at `https://github.com/Fomalhaut647/Replication-Shrinking-the-Cross-Section`.

E Division of labor

This is Group 3; the members and the division of labor are as follows.

Table 3: Group members and division of labor (Group 3)

Name (ID)	Major	Main responsibilities
Liu Zhiqi (2300012860)	Artificial Intelligence	Programming and implementation: all code (the unified numerical core, the four core figures, and the five extensions), the figures, the one-click reproduction pipeline, and the self-tests.
Mao Zhiyuan (2300012108)	Mathematics and Applied Mathematics	Theoretical derivations: the shrinkage prior, the general-η posterior and shrinkage weight, and the level-versus-relative-shrinkage argument.

The following were completed jointly by both members: the literature review of the original paper, the selection of the extension directions, and the writing of this report.